

Numerical Analysis I - Notes

1 Numbers, Problems and algorithms

We are more interested in getting the relative error than the absolute error. The relative error is defined as

$$\frac{|x - x_{\text{true}}|}{|x_{\text{true}}|}$$

where x is the computed value and x_{true} is the true value.

WE can use conditional numbers to determine how good our computation is. The conditional number is defined as

$$\kappa = \frac{\|J\| \cdot \|x\|}{\|f(x)\|}$$

where J is the Jacobian matrix of the function f .

1.1 Horner's algorithm

$$f(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$$

can be evaluated using Horner's algorithm:

$$f(x) = a_0 + x(a_1 + x(a_2 + \cdots + x(a_{n-1} + xa_n) \cdots))$$